

Team Worksheet — Lab 21

Analyze Tollgate — Proving Root Cause and Telling the Data Story

Team members: _____

Date: _____

1. Our understanding of the problem

2. Our working

Use the data in `lab-21-analyze-tollgate-evidence.xlsx`.

3. What we produced

Deliverable: An Analyze tollgate pack with proven root causes, evidence and a business-language summary.

4. What the evidence does NOT tell us

5. Our conclusion

6. Step notes

Step 1. Assemble every analytical result from Labs 15-20: stratification, hypothesis tests, regression model, ANOVA, multi-vari and chi-square.

Step 2. For each candidate cause, record the verdict — PROVEN (statistically supported), REJECTED (tested and cleared) or UNTESTED (no data available).

Step 3. Apply the triangulation rule: a root cause is strongest when supported by more than one independent method — for instance stratification plus a hypothesis test plus a regression coefficient.

Step 4. Check each proven cause for practical significance. Quantify how much of the Y gap each explains, in percent and in dollars.

Step 5. Rank your proven causes by the size of their contribution to the Y gap.

Step 6. Confirm your causes are ACTIONABLE — a proven cause you cannot influence belongs in the risk register, not the improvement plan.

Step 7. Translate each finding into one business sentence: 'Night shift at Suzhou runs 2.3 percentage points lower first-pass yield, costing about \$180k annually.'

Step 8. Build the tollgate pack: baseline, analysis method, proven causes, rejected causes, quantified impact and the recommendation to proceed.

Step 9. Rehearse the challenges you should expect: was the sample adequate, were assumptions checked, could a confounding variable explain this, is the measurement system valid?

Step 10. Prepare your evidence for each challenge — this is exactly what the Day 5 steering committee will probe.

Step 11. Present the tollgate to a peer group and capture their challenges, then strengthen any weak points before Day 4.

7. Self-check

Every candidate cause carries a verdict, proven causes are triangulated across multiple methods and quantified in dollars, and you have rehearsed a prepared response to each expected challenge.

Complete? ■ Yes ■ Not yet

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